

Kinga Tshering

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🌐 Personal Website • 🎓 Google Scholar • 🐙 GitHub • 🔗 LinkedIn

RESEARCH INTERESTS

Evaluation and reliability of large language models • efficient and resource-aware deep learning, including network pruning and model compression • interpretability of learned representations.

EDUCATION

North South University

2022 – 2026

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

- CGPA 3.90 / 4.00 • **Summa Cum Laude**
- **His Majesty's Scholarship** — awarded nationally to top Bhutanese students

RESEARCH EXPERIENCE

Research Intern — University of Arizona

2026 – Present

Remote

College of Information Science • Advisor: Prof. Haw-Shiuan Chang

- Extending Asymptotic Probability Decoding (Chang et al., 2024) from extrapolation across model *scale* to extrapolation across *context length*, removing the method's reliance on a family of same-recipe language models at multiple sizes.
- Auditing the training pipeline — curve-fitting objective, top-*k* candidate selection, and numerical conditioning of the parametric decay fit — against the original formulation to isolate sources of training instability.

Undergraduate Researcher — North South University

2024 – 2026

Dhaka, Bangladesh

Department of ECE • Advisor: Prof. Shafin Rahman

- Studied sparse subnetworks in hyperspectral image classification under the Lottery Ticket Hypothesis, evaluating iterative magnitude pruning across 3D CNN, hybrid 3D–2D CNN, and Mamba backbones on three benchmark scenes.
- Designed evaluation protocols for post-training language model assessment, separating instruction-following compliance from answer correctness across arithmetic, logical, and code-generation benchmarks.
- Built benchmark construction and automated evaluation pipelines for formal-language tasks spanning the Chomsky hierarchy.

PUBLICATIONS AND PREPRINTS

Own name in **bold**. * denotes equal contribution.

- 1 TimeSpot: Benchmarking Geo-Temporal Understanding in Vision–Language Models in Real-World Settings
Azmine Toushik Wasi, Shahriyar Zaman Ridoy, Koushik Ahamed Tonmoy, **Kinga Tshering**, S. M. Muhtasimul Hasan, Wahid Faisal, Tasnim Mohiuddin, Md Rizwan Parvez
International Conference on Machine Learning (ICML), 2026 • Accepted
- 2 From Language Specifications to Executable Turing Machines: Evaluating LLMs as Computational Machine Designers
Shahriyar Zaman Ridoy, S. M. Muhtasimul Hasan, Azmine Toushik Wasi, **Kinga Tshering**, Koushik Ahamed Tonmoy, Md Mosaddek Khan
Findings of the ACL: EMNLP, 2026 • Accepted

- 3 The Screen-Time Trap: An Explainable Machine Learning Framework for Pediatric Screen-Time Overuse Prediction
Kinga Tshering, Mushfika Hossain Piya, Ashraf Sharif, Marzan Islam
Zorig Melong, Vol. 9, No. 1, 2026
- 4 Black Box to Survivor Bands: Decoding Sparse Subnetworks in Hyperspectral Image Classification
Omar Sadat*, Shahriar Kamal*, Kinga Tshering*, Plabon Sarkar*
Transactions on Machine Learning Research (TMLR) • Under review
- 5 H-DAFON: Hierarchical Dynamic Attribute-Guided Network for Few-Shot Open-Set Skin Lesion Recognition
Kinga Tshering, Osama Nadeem, S. M. Samiul Islam, Taufiq Elahi, Riasat Khan
Systems and Soft Computing • Under review
- 6 Beyond Strict Accuracy: A Compliance–Correctness Decomposition for Post-Training Evaluation of Small Language Models
Kinga Tshering, Sumon Das
In preparation

TEACHING

Undergraduate Teaching Assistant

2024 – 2025

Dhaka, Bangladesh

Department of ECE, North South University

- CSE 215 — Programming Language II (Java) • Prof. Rajesh Palit, Prof. Shafin Rahman, Md. Ishan Arefin Hossain
- CSE 332 — Computer Architecture and Organization • Prof. Nabil Bin Hannan
- Held weekly office hours, graded assignments and examinations, supported laboratory sessions

SELECTED SOFTWARE

DrukAgriLink

2025

druk-agri-link.vercel.app

Agricultural coordination platform for Bhutanese smallholder farmers

- Deployed a multi-role platform (farmers, buyers, coordinators, transporters) implementing a staged workflow from harvest listing through demand matching to fulfillment, with role-based access enforced by PostgreSQL row-level security.
- Integrated a 1,051-record Bhutanese administrative hierarchy (Dzongkhag–Gewog–Chiwog) into a cascading selector for standardized geographic entry.
- Next.js, TypeScript, Supabase (PostgreSQL, Auth), Tailwind CSS; deployed on Vercel.

IntelliExpense

2024

Course project

Machine learning system for personal expenditure analysis

- Expense classification, budget forecasting, and spending-pattern analysis over user transaction histories • Django, scikit-learn, Pandas, NLTK.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, TypeScript, SQL
ML Stack	PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn, Matplotlib
Research Infra.	SLURM / HPC clusters, Git, L ^A T _E X, Weights & Biases
Web	Next.js, Django, Supabase / PostgreSQL

HONORS AND AWARDS

- His Majesty's Scholarship, Government of Bhutan 2022 – 2026
- Summa Cum Laude, North South University 2026

- **Academic Topper Certificates**, awarded consecutively by His Majesty the King of Bhutan, Grades 6–11 2011 – 2016
- **Best Student of the Year**, Dechentsemo Central School 2016
- **5th Place**, 9th Golden Youth Award Camp, Bhutan 2016

LEADERSHIP AND SERVICE

- **Coordinator**, Bhutanese Students Association, North South University 2022 – Present
 - **Volunteer President**, CSO Nazhoen Lamtoen, Bhutan 2020 – 2021
- Supported children in conflict with the law alongside the Regional Case Management Officer ▪ organized a community campaign in Phuentsholing for World Mental Health Day ▪ HeForShe campaign advocate in Bhutanese schools.

LANGUAGES

Dzongkha (native) ▪ English (fluent) ▪ Nepali (conversational) ▪ Hindi (conversational) ▪ Bangla (elementary)

REFERENCES

Dr. Shafin Rahman
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Faculty Profile

Dr. Rajesh Palit
Professor
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Dr. Shams Forruque Ahmed
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